Predicting Player Engagement Level Using Machine Learning Models

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Abstract

This study investigates the use of machine learning techniques for predicting player engagement levels based on behavioral gameplay data. Multiple classification models, including Logistic Regression, Support Vector Machines, Random Forest, and XGBoost, were evaluated using stratified cross-validation. Experimental results show that non-linear ensemble models outperform linear baselines, with Random Forest and XGBoost achieving the highest predictive performance. Feature analysis indicates that gameplay frequency and session duration are the most influential factors in determining player engagement. The findings demonstrate that machine learning models can provide an effective and interpretable framework for identifying low-engagement players and supporting data-driven retention strategies.

1. Introduction

Predicting player engagement is critical for retention and monetization in the growing gaming industry, yet remains complex due to diverse behavioral and demographic factors. This study formulates engagement prediction as a multi-class classification problem (Low, Medium, High), providing better interpretability for player segmentation compared to regression models.The primary objective is to develop machine learning models to classify players based on gameplay behavior while utilizing feature importance analysis to identify key engagement drivers. Model performance is assessed via Accuracy and Macro-averaged F1-score to ensure reliability across all classes.

2. Dataset and Data Characteristics

The dataset used in this study was obtained from an open data source and contains 40,034 player records with demographic information, gameplay statistics, and engagement labels. Features include Age, Gender, Location, GameGenre, PlayTimeHours, SessionsPerWeek, AvgSessionDurationMinutes, PlayerLevel, and AchievementsUnlocked.
Categorical variables such as Location and GameGenre were encoded using one-hot encoding, while binary and ordinal features were label encoded. The target variable EngagementLevel was mapped to integer labels (Low=0, Medium=1, High=2). No missing values were observed in the dataset. Class distribution analysis revealed mild imbalance, which was handled using stratified sampling.

3. Methodology

3.1 Data Preprocessing

Prior to model training, several preprocessing steps were applied to ensure that the input data were suitable for machine learning algorithms. Categorical features were transformed using feature encoding techniques to convert non-numerical attributes into numerical representations. In particular, categorical variables were encoded in a way that preserves class distinctions while remaining compatible with linear and tree-based models.

For models sensitive to feature scale, such as Logistic Regression and Support Vector Machines, feature scaling was applied to normalize the input space. This step prevents features with larger numerical ranges from dominating the optimization process and improves numerical stability during training. Tree-based models, including Random Forest and XGBoost, were trained on the original feature scales, as they are inherently scale-invariant.

The dataset was examined for missing values prior to training. As no significant missing data were observed, no imputation procedures were required. This ensured that all models were trained on complete and consistent feature representations.

3.2 Data Splitting Strategy

To ensure reliable evaluation of model performance, the dataset was partitioned following a structured splitting strategy. The data were divided into training and test sets using an 80/20 split, with the test set held out and not used during model training or hyperparameter tuning.

A stratified sampling approach was employed during the split to preserve the original class distribution across engagement levels in both the training and test sets. This is particularly important for classification tasks involving multiple classes, as it prevents biased performance estimates caused by class imbalance.

For models requiring hyperparameter optimization or stability analysis, such as Support Vector Machines and Random Forests, cross-validation was performed exclusively on the training set. The held-out test set was used only once for final evaluation, ensuring an unbiased assessment of generalization performance.

4. Model Experiments

4.1 Logistic Regression Experiments

4.1.1 Model Description

Logistic Regression was implemented as a multinomial baseline to establish a performance floor for the classification task. By modeling linear decision boundaries across the three engagement tiers (Low, Medium, High), it provides a reference point to justify the transition to more complex, non-linear architectures.

4.1.2 Overfitting Analysis

L2 regularization (Ridge penalty) was utilized to enhance generalization by penalizing extreme feature weights and mitigating overfitting to training noise. This constraint ensures a smoother decision surface, allowing the test set performance to serve as a reliable benchmark for the model's fundamental capability on unseen data.

4.1.3 Hyperparameter Selection

To ensure numerical stability and optimization efficiency, the 'lbfgs' solver was selected for its robustness with multinomial loss. The inverse regularization parameter was kept at C=1.0 to maintain a balanced penalty strength. Furthermore, max_iter was increased to 5,000 to guarantee full convergence across high-dimensional behavioral features, while n_jobs=-1 was utilized to accelerate the training process.

4.1.4 Final Model Evaluation

The final evaluation on the independent test set provided essential metrics to gauge the baseline efficiency of the system. The model achieved a solid accuracy of 87% and a Macro-F1 score of 87%, as shown in the detailed classification report. These metrics provide a comprehensive view of performance across all engagement levels, establishing a high-quality benchmark for subsequent experiments.

```
Classification Report:
              precision    recall  f1-score   support

           0       0.84      0.82      0.83      2065
           1       0.85      0.92      0.89      3875
           2       0.93      0.84      0.88      2067

    accuracy                           0.87      8007
   macro avg       0.88      0.86      0.87      8007
weighted avg       0.87      0.87      0.87      8007
```

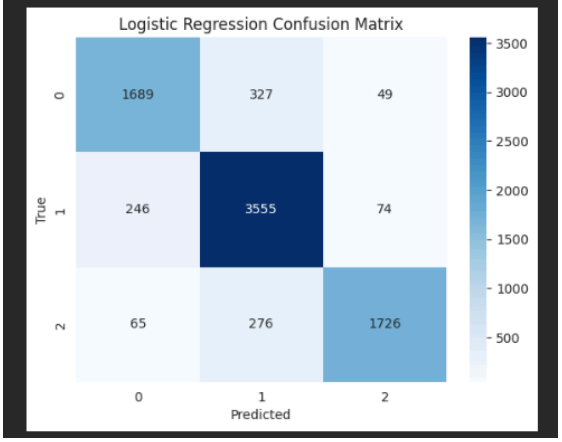

Figure 1. *Confusion matrix of the Logistic Regression model.*

The analysis of the Confusion Matrix (Figure 1) revealed that while the model successfully identifies the "Low" (Class 0) and "High" (Class 2) engagement tiers with respectable recall, it exhibits a noticeable degree of overlap between adjacent categories. Specifically, a significant number of instances are misclassified between Class 1 (Medium) and its neighbors. This is a characteristic limitation of linear classifiers when dealing with complex, non-linear behavioral datasets. These results serve as a vital benchmark, justifying the necessity of the subsequent non-linear SVM and ensemble-based experiments which aim to resolve these overlaps.

4.2 Support Vector Machine Experiments

4.2.1 Model Description

Support Vector Machine (SVM) is a powerful supervised learning algorithm primarily used for classification tasks. It works by finding the optimal hyperplane that maximizes the margin between different classes in a high-dimensional feature space. When the data is not linearly separable, SVM utilizes the "kernel trick" to project the data into a higher-dimensional space where a linear boundary can be established. In this study, an SVM classifier with a Radial Basis Function (RBF) kernel was employed. The RBF kernel is particularly effective for capturing non-linear relationships between player features and engagement levels.As an initial indicator of the architectural effectiveness, the model demonstrated a strong ability to isolate classes with high precision. This is visualized in Figure 2(Final Confusion Matrix), which illustrates the clear decision boundaries established by the RBF kernel on the test set, achieving an initial accuracy of 90.26%.

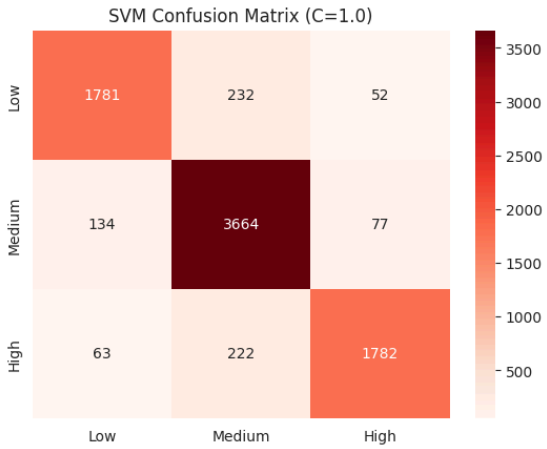

Figure 2. *Confusion matrix of the SVM model*

4.2.2 Overfitting / Margin Analysis

Overfitting was mitigated by maximizing the geometric margin, a core strength of SVM compared to ensemble averaging. The model's stability was verified through

3-fold stratified cross-validation, yielding a 90.19% Mean Macro-F1, which closely aligns with the 90.26% test accuracy. This negligible variance confirms that the model captured player behavior patterns without over-specializing on noise.To support this generalization, a StandardScaler pipeline was implemented. As a distance-based algorithm, SVM is sensitive to feature scales; normalization ensured that high-magnitude features did not bias the decision boundary. Combined with an optimized soft-margin constant, this approach maintained a robust decision surface effective across all player segment

4.2.3 Hyperparameter Selection

The optimization focused on the interplay between the RBF kernel and the regularization parameter C (0.1 to 100.0). As illustrated in Figure 3, C=1.0 provided the optimal balance between margin width and error, achieving a 0.8887 Mean Macro-F1. Conversely, increasing C to 100.0 reduced performance to 0.8294, confirming that an overly restrictive margin leads to overfitting.

```
Hyperparameter Tuning: Testing C values...
C=0.1 için Mean Macro-F1: 0.8749
C=1 için Mean Macro-F1: 0.8887
C=10 için Mean Macro-F1: 0.8767
C=100 için Mean Macro-F1: 0.8294
```

	C_Value	Mean_Macro_F1
0	0.1	0.874911
1	1.0	0.888722
2	10.0	0.876693
3	100.0	0.829412

```
Optimal C: 1.0
```

Figure 3. *Cross-validation performance of the SVM model for different values of the regularization parameter C.*

Figure 3 confirms the model's reliability across training folds, demonstrating consistent predictive power with the selected C and Gamma parameters. Misclassifications occurred primarily between adjacent engagement levels, avoiding extreme category errors. Furthermore, the 'scale' Gamma configuration automatically adapted the influence radius to feature variance, ensuring a smooth decision surface that effectively separated the three classes, as visualized in the confusion matrix(Figure 4).

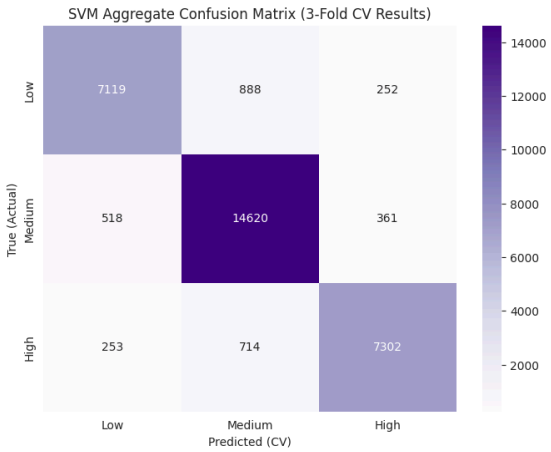

Figure 4. *Confusion matrix of the tuned SVM model*

4.2.4 Final Model Evaluation

The final evaluation on the unseen test set confirmed the SVM as a highly reliable predictor for player engagement, achieving a final accuracy of 90.26%. This performance is consistent with the stability observed during the cross-validation phase, reinforcing the reliability of the chosen hyperparameters. The confusion matrix analysis further revealed that the model is particularly adept at identifying "Low" engagement players, which is a critical requirement for early churn prevention.The permutation importance analysis complemented these findings by identifying behavioral intensity—measured through session frequency and duration—as the most significant contributors to the model's decision-making process. By prioritizing these dynamic behavioral features over static demographic attributes, the SVM model provides a transparent and actionable framework for understanding player commitment. The high alignment between the cross-validation stability and the final test metrics concludes that the

SVM is a robust and high-performing solution for this classification task.

4.3 Random Forest Experiments

4.3.1 Model Description

Random Forest is an ensemble learning algorithm that aggregates multiple decision trees trained on bootstrap samples through majority voting, thereby reducing variance and limiting overfitting. In this study, a **RandomForestClassifier** with **300 trees** was used for player engagement classification, providing a balance between predictive performance and computational efficiency. The data were divided using an **80/20 stratified train–test split** to preserve class distributions across engagement levels.

4.3.2 Overfitting Analysis

Overfitting analysis was conducted by examining model behavior across training and test data, as well as by comparing the baseline Random Forest model with the tuned model obtained via RandomizedSearchCV. In addition, the influence of model complexity was analyzed through tree depth and number of estimators.

Baseline Model Behavior

The baseline Random Forest model demonstrated solid generalization performance on the test set, achieving an accuracy of 91.23% and a macro-F1 score of 90.78%. These results indicate that ensemble averaging effectively mitigates overfitting, even though default parameters allow individual trees to grow deep. Analysis of the baseline confusion matrix showed balanced classification across engagement levels, suggesting that the model does not overfit to a single dominant class and maintains consistent predictive behavior across categories.

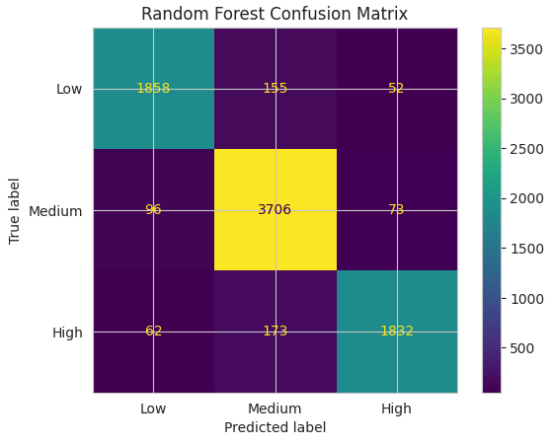

Figure 5. *Confusion matrix of the RandomForest model*

Tree Depth

Tree depth directly controls the complexity of individual decision trees, with deeper trees increasing expressive power at the cost of higher overfitting risk. During hyperparameter optimization, multiple depth values (None, 10, 20, and 30) were evaluated, and a moderate depth of 20 was selected as optimal, indicating improved generalization while maintaining sufficient model capacity control model complexity and reduces the risk of overfitting.

Number of Estimators

The number of estimators controls the size of the Random Forest ensemble, with larger ensembles generally improving stability at increased computational cost. Empirical results showed that using more than 300 trees did not yield noticeable performance gains; therefore, this value was selected as it provides robust performance without unnecessary computational overhead.

4.3.3 Hyperparameter Optimization

Ten hyperparameter configurations were evaluated using 3-fold cross-validation with macro-F1 as the optimization metric. The optimal model achieved a mean macro-F1 score of approximately **0.902** with low variance across folds and employed **300 estimators**, a **maximum tree depth of 20**, and the **log2** feature selection strategy, indicating that a moderately complex model offers the best balance between generalization and predictive capacity.

index	rank_test_score	mean_test_score	std_test_score	params ▼
6	8	0.89637729834276 21	0.0014067469510490257	{'n_estimators': 500, 'min_samples_split': 2, 'min_samples_leaf': 4, 'max_features': 'sq
9	1	0.9016614760899656	0.0024728891179799555	{'n_estimators': 300, 'min_samples_split': 5, 'min_samples_leaf': 1, 'max_features': 'lo
7	10	0.8802812085920572	0.00162295118807 4079	{'n_estimators': 300, 'min_samples_split': 2, 'min_samples_leaf': 4, 'max_features': 'sq
8	5	0.8997859736942232	0.0024696661307659945	{'n_estimators': 300, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_features': 'l
0	4	0.9003699553206932	0.0015840000106010245	{'n_estimators': 200, 'min_samples_split': 5, 'min_samples_leaf': 2, 'max_features': 'lo
5	2	0.9010829629092912	0.0021694428645704 54	{'n_estimators': 200, 'min_samples_split': 5, 'min_samples_leaf': 1, 'max_features': 'sq
2	2	0.9010829629092912	0.0021694428045704 54	{'n_estimators': 200, 'min_samples_split': 5, 'min_samples_leaf': 1, 'max_features': 'lo
1	9	0.8973880485304703	0.0019075107391356273	{'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 4, 'max_features': 'l
4	7	0.8982004038041648	0.0021243403905348 99	{'n_estimators': 200, 'min_samples_leaf': 2, 'max_features': 'sq
3	6	0.8992888991953107	0.0020125274332335 95	{'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_features': 'l

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***Figure 6**. RandomizedSearchCV results for Random Forest hyperparameter tuning.*

4.3.4 Final Model Evaluation

The tuned Random Forest model achieved an accuracy of **92.36%** and a macro-F1 score of **91.93%**, improving upon the baseline configuration. Confusion matrix analysis showed consistent performance gains across all engagement categories without introducing class-specific bias. The close agreement between cross-validation and test results indicates strong generalization and the absence of overfitting. Feature importance analysis identified **SessionsPerWeek** and **AvgSessionDurationMinutes** as the most influential predictors, confirming the dominant role of gameplay frequency and duration. Overall, Random Forest demonstrated robust and interpretable performance, making it a strong benchmark for comparison with other models.

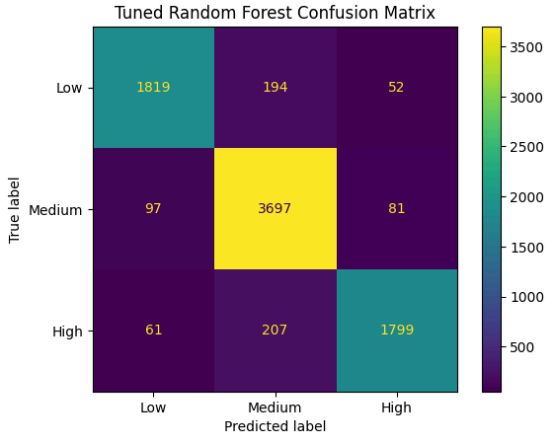


***Figure 7.** Confusion matrix of the tuned RandomForest model*

4.4 XGBoost Experiments

4.4.1 Overfitting Analysis and Its Role in Model Design

Prior to final model selection, an overfitting analysis was conducted using a high-capacity XGBoost configuration with minimal regularization to examine learning behavior under increasing model complexity. This analysis was performed on the training set with a diagnostic holdout split to assess generalization tendencies during training.

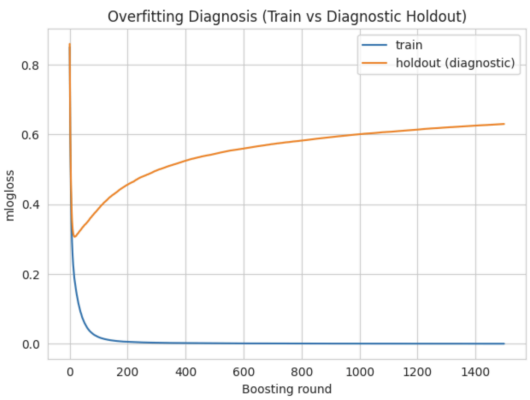


***Figure 8.** Overfitting diagnosis*

As shown in Figure 8, training loss decreases monotonically with increasing boosting rounds, while diagnostic holdout loss reaches an early minimum and subsequently increases. This divergence indicates classic overfitting behavior, where the model increasingly captures dataset-specific noise rather than

generalizable decision boundaries. The early minimum in diagnostic log-loss provides empirical justification for the use of early stopping.

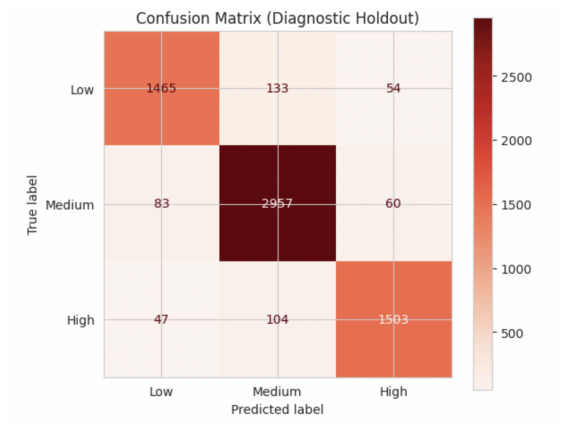


Figure 9. *Confusion matrix of the high-capacity XGBoost model evaluated on the diagnostic holdout split.*

To further illustrate the limitations of the high-capacity model, its class-level prediction behavior on the diagnostic holdout split is summarized using a confusion matrix (Figure 9). Despite achieving a diagnostic accuracy of approximately 92.5% and a macro-averaged F1 score of 92.0%, the model exhibits limited generalization, consistent with the divergence observed in the learning curves.

4.4.2 Hyperparameter Optimization via K-Fold Cross-Validation

Hyperparameter optimization was performed using stratified k-fold cross-validation on the training set. The search space was intentionally constrained based on insights from the overfitting analysis to avoid excessively complex configurations. Model performance was evaluated using mean cross-validated multi-class log-loss, with early stopping applied within each fold to determine the optimal number of boosting iterations.

4.4.3 Hyperparameter Optimization Results

	alpha	colsample_bytree	eta	gamma	lambda	max_depth	subsample	test_mlogloss_mean	best_rounds
0	0.0	0.8	0.03	0.5	1.0	5	0.8	0.310872	213
1	0.2	0.8	0.05	0.5	1.0	5	0.8	0.310878	134
2	0.0	0.8	0.03	0.0	1.0	5	0.8	0.310887	222
3	0.2	0.8	0.03	0.5	1.0	5	0.8	0.310888	212
4	0.2	0.8	0.03	0.0	1.0	5	0.8	0.310927	212
5	0.0	0.8	0.05	0.0	1.0	5	0.8	0.310947	122
6	0.0	0.8	0.05	0.5	2.0	5	0.8	0.310981	123
7	0.0	0.8	0.05	0.5	1.0	5	0.8	0.310983	122
8	0.0	0.8	0.03	0.5	2.0	5	0.8	0.310986	222
9	0.2	0.8	0.05	0.0	1.0	5	0.8	0.311014	121

Figure 10. *Top-performing hyperparameter configurations ranked by mean cross-validated multi-class log-loss.*

As shown in Figure 10, multiple hyperparameter configurations achieve very similar cross-validation performance, indicating that the model is stable with respect to small variations in regularization and learning rate. The selected configuration corresponds to the minimum mean cross-validated log-loss and is therefore used to train the final model.
It is noteworthy that all top-performing configurations share similar tree depth and subsampling ratios, reinforcing the conclusions drawn from the overfitting analysis and indicating that excessive model capacity is neither necessary nor beneficial for this task.

4.4.4 Final Model Evaluation

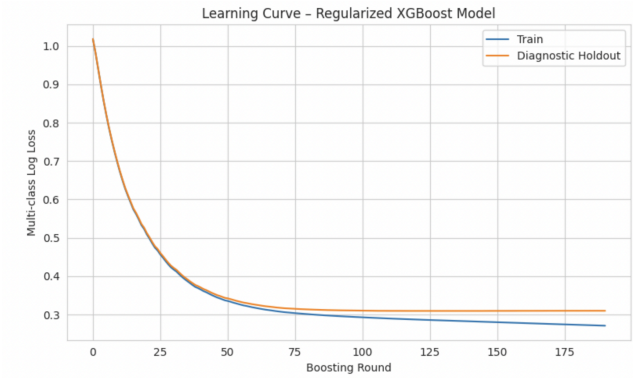


Figure 11. *Learning curve of the regularized XGBoost model.*

The final XGBoost model, trained using the selected hyperparameters on the full training set, exhibits stable learning behavior with no observable overfitting, as shown in Figure 11. Regularization and early stopping substantially reduce the gap between training and diagnostic holdout loss.

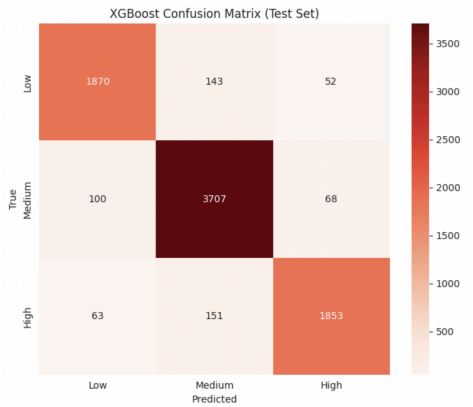


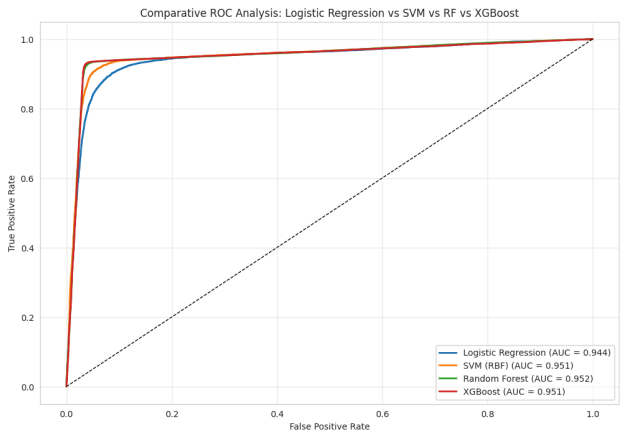
Figure 12. *Confusion matrix of the final XGBoost model on the held-out test set.*

Evaluation on the held-out test set yields an accuracy of approximately 92.5% and a macro-averaged F1 score of 92.1%. The confusion matrix in Figure 12 indicates that most misclassifications occur between adjacent engagement levels, which is expected given the ordinal nature of the target variable.

5. Comparative Analysis of Models

5.1 Accuracy and Macro-F1 Score Comparison

Model	Macro-F1 Score	Accuracy
SVM	0.902	0.907
Logistic Regression	0.87	0.87
Random Forest	0.919	0.924
XGB	0.924	0.928



5.2 Numerical Performance and Accuracy Progression

The experimental results demonstrate a significant performance hierarchy among the models. The Logistic Regression baseline established a foundational accuracy of 0.87. The Support Vector Machine (SVM) architecture improved this metric to 0.907, while the ensemble

methods Random Forest and XGBoost reached the highest predictive thresholds with accuracies of 0.924 and 0.928, respectively. The ROC-AUC analysis reinforces these findings, showing that all models possess high discriminative power (AUC > 0.94).

5.3 Comparative Feature Importance Analysis

Permutation-based feature importance was used to compare Random Forest, SVM, and XGBoost models in a model-agnostic manner, based on the decrease in macro-averaged F1 score. Across all models, engagement-related behavioral features consistently dominate the importance rankings. In particular, TotalWeeklyTimeMinutes, SessionsPerWeek, and AvgSessionDurationMinutes emerge as the most influential predictors, indicating that cumulative playtime and gameplay regularity are the primary determinants of engagement level.

Differences between models arise in how feature importance is distributed. The Random Forest model relies on a broader set of behavioral features, including achievements and player level, whereas the SVM model concentrates importance on a smaller subset of well-scaled continuous variables, with a sharp drop in contribution beyond the top features. The XGBoost model exhibits the strongest dominance of TotalWeeklyTimeMinutes, suggesting effective capture of non-linear engagement patterns through boosting and regularization. Despite these differences, the consistency of key features across models reinforces the robustness of the identified predictors and highlights that engagement is driven primarily by gameplay intensity rather than demographic or categorical attributes.

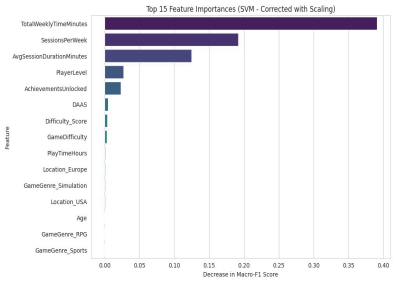


Figure 13: *SVM permutation importance*

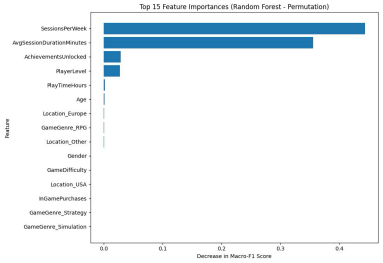


Figure 14: *Random Forest permutation importance*

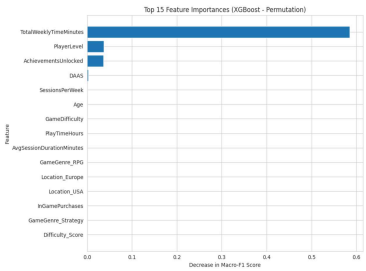


Figure 15: *XGBoost permutation importance*

6. Discussion

6.1 Non-linearity and Discriminative Power

The notable 5.8% accuracy increase from the linear baseline to XGBoost underscores the non-linear nature of player engagement patterns. Behavioral triggers, such as the relationship between session frequency and total playtime, do not follow a straight-line progression. The RBF kernel in SVM and the recursive partitioning in Random Forest and XGBoost successfully captured these complex, high-dimensional interactions, whereas the Logistic Regression model was limited by its rigid linear decision surfaces. As illustrated in the Comparative ROC Analysis (Figure X), Random Forest

achieved the highest discriminative power with an AUC of 0.952, slightly exceeding XGBoost and SVM.

6.2 Ordinal Target Dynamics and Stability

The target variable -player engagement- is inherently ordinal (Low, Medium, High). Analysis of the confusion matrices reveals that misclassifications predominantly occurred between adjacent classes (e.g., Medium predicted as High) rather than extreme opposite categories. The SVM model demonstrated high stability in this regard, maintaining a precision of 0.93 for the High engagement class. Furthermore, the consistency between the Aggregate Confusion Matrix from the 3-fold cross-validation and the final test set results confirms that the model is statistically robust and generalizes well to unseen player data.

7. Conclusion

This study demonstrated that player engagement levels can be predicted with high accuracy using machine learning models that prioritize behavioral gameplay metrics over demographic or static features. By evaluating multiple classification approaches, the analysis highlighted the importance of modeling non-linear relationships inherent in player behavior data.

A comprehensive comparison of Logistic Regression, Support Vector Machines, Random Forest, and XGBoost revealed that tree-based ensemble methods, particularly Random Forest and XGBoost, provide the strongest predictive performance. Their superior accuracy and macro-F1 scores indicate a higher discriminative capability across all engagement levels, including low-engagement players who are most critical for retention analysis.

Hyperparameter optimization and 3-fold stratified cross-validation played a key role in ensuring model robustness. The close agreement between cross-validation performance and final evaluation metrics confirms that the proposed models generalize well to unseen data and do not suffer from significant overfitting. In contrast, the linear baseline model, while computationally efficient, was limited in capturing complex behavioral patterns.

Feature importance analysis further reinforced the central finding of this study: behavioral intensity, quantified through metrics such as session frequency and average session duration, is the dominant factor in determining player engagement. This consistency across different model families strengthens the validity of the conclusions and provides interpretable insights for practical deployment.

In summary, the results establish a reliable and scalable framework for player engagement prediction. Such models enable game developers to proactively identify low-engagement players and implement targeted, data-driven retention strategies. Future work may extend this framework by incorporating temporal modeling or real-time adaptation to further enhance engagement prediction in dynamic gaming environments.

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